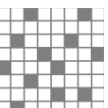


Release Notes

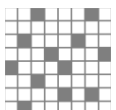
ALP-4.3 API



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ALP-4.3 API Release Notes

Introduction

This document summarizes release 23 of the ALP-4.3 application programming interface (API). The following remarks are to indicate additional programming capabilities.

These release notes apply to ALP-4.3 components of the following versions:

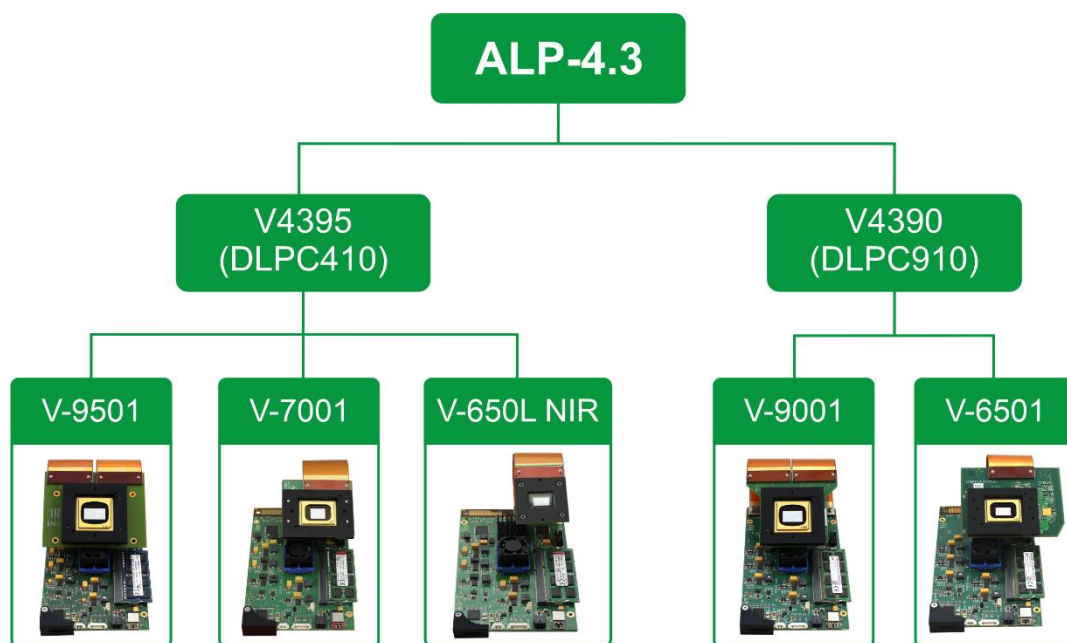
- API (alp4395.dll): 1.0.23.77
- API header file (alp.h): 27
- ALP API description (ALP-4.3 API description.pdf): 2023-09-28

ALP APIs are compatible to each other regarding their software interface. However, they are specialized to support individual hardware versions. The ALP hardware/software table summarizes the combinations for the three versions of DLL:

- **ALP-4.1** (alpD41.dll)
- **ALP-4.2** (alpV42.dll)
- **ALP-4.3** (alp4395.dll)
- **ALP-4.4** (alp44.dll)

ALP hardware/software table:

DLP DMD Chip	DLP Controller FPGA	DLP Controller Board	Supporting ALP-4 Software	ViALUX Product
DLP650L DLP7000 DLP9500	DLPC410	ViALUX VX4100 or TI DCB4100	ALP-4.1 (as an option)	Discovery™4100 Development Kit
DLP7000		ViALUX V4100	ALP-4.2	V-7000
DLP650L		ViALUX V4395	ALP-4.3	V-650L
DLP7000				V-7001
DLP9500				V-9501
DLP6500	DLPC910	ViALUX V4390		V-6501
DLP9000X				V-9001
DLP7000	DLPC410	ViALUX VDIF	ALP-4.4	V-7001+



Summary of Modifications

New V-Module V-9001 turned

Release 23 of supports new types of V-Modules:

- V-9001t VIS (turned) : DLP9000X mounted on a turned DMD extension board (turn angle is 45°)
- V-9001t UV (turned)

New ALP extension: DYN_SYNC_OUT

A new ALP API extension called DYN_SYNC_OUT is now available. The extension provides a special variant of the default frame trigger. This trigger can be configured to be active for multiple consecutive frames and to have a periodic repetition. Please contact ViALUX for additional information on this extension.

New ALP extension: Row based bit plane lookup table

This extension implements a row based bit plane lookup table, which enables the customer to display different bit planes (up to 32) on the dmd simultaneously.

Support of LED Type CBM-120-FR

The CBM-120-FR LED is now supported by the ALP-4.3 API with the new ALP LedType ALP_HLD_CBM120_FR.

Additional ALP Samples

Two new samples called “ALP ImageConverter Sample” and “ALP Streaming Sample” are provided with this release. The Image Converter Sample will take image files in common file formats (like .png or .bmp) and convert them into the proprietary format, used by the ALP API. The Streaming Sample will read in the converted image data and transfer it to the V-Module while already transferred data is projected on the DMD simultaneously. This way a “streaming like” behavior is achieved.

The two samples are meant to be used in combination with each other. An example of how to call the two programs is provided with the “Convert and Streaming Sample.bat” located in the Samples subfolder of the installation.

Bug-fix

An issue has been fixed, where SynchPulses with very long SynchPulseWidths (more than 8 seconds) were not outputted with the correct length.

An issue was fixed where, in rare cases, concurrent calls of AlpSeqPut and other ALP functions could lead to a permanent deadlock of the ALP API.

Deployment

ALP-4.3 Release 23 is an update to the API library alp4395.dll. Existing installations need only a replacement of this DLL file. No driver update or re-installation is required.